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Google Patents API

API uptime 99.952%

Google Patents API allows you to scrape patent and scholar results from Google Patents.

The API endpoint is `https://serpapi.com/search?engine=google_patents`

Head to the playground for a live and interactive demo.

q Search Query

solar panel

Search

Patents solar panel

About 118864 results

Sort by · Relevance · Group by · None · Deduplicate by · Family · Results / page · 10

Apparatus for two-way tracing and concentrating sunlight of roof installation ...
EP US CN JP KR · EP2264377A2 · Young Hwan Park · Green Plus Co. Ltd.
Priority 2009-06-03 · Filed 2009-12-04 · Granted · Published 2010-12-22
a plurality of second open push rods which have one sides hinge-coupled to the plurality of se pipes side by side to move horizontally, and the other sides hinge-coupled to the lower portion plurality of second **solar module** plates respectively to lift the second **solar module** ...

Photovoltaic support
CN NL · CN115498949A · 赵海玉 · 华能新能源股份有限公司河北分公司
Priority 2022-09-27 · Filed 2022-09-27 · Granted · Published 2022-12-20
Background technique In a photovoltaic power generation system, the brackets used to suppo modules are generally divided into fixed brackets and tracking brackets. Among them, tracking further divided into horizontal single-axis tracking brackets, oblique single-axis ...

Electricity production blocking shield window using existing technology
KR · KR20140002094U · 백창화 · 백창화
Priority 2012-10-02 · Filed 2012-10-02 · Granted · Published 2014-04-14
[Fig.4] [Front view] Horizontal louver type solar collector [Fig.-2] The drawing from the front of becomes a flat panel when all of the **solar module** panels are folded to the bottom. Doors with window) ① Frame on the solar collector body ② **Panel-foldable solar module** ③ ...

Linkage type photovoltaic power generation sunshade insulation integrated ...
CN · CN109972776B · 张玉坤 · 天津大学
Priority 2019-03-27 · Filed 2019-03-27 · Granted 2024-04-09 · Published 2024-04-09
The invention aims to overcome the defects of the prior art, and provides the linkage type photovoltaic powe sunshade insulation integrated double-layer glass curtain wall, which can utilize **solar energy** to carry out m production, can meet the requirements of building ...

Sunlight intelligent tracking heat collection system and method based on **sola**
CN · CN112984831A · 张月 · 张月
Priority 2021-03-30 · Filed 2021-03-30 · Granted · Published 2021-06-18
1. The intelligent sunlight tracking and heat collecting system based on **solar energy** comprise plate (1), a **solar panel** (2) and a ray tracking sensor (3), and is characterized in that a contract mechanism (4) is fixed on the upper surface of the mounting plate (1), a ...

Double-sided battery pack
CN · CN102254977A · 王兴华 · 泰通(泰州)工业有限公司
Priority 2010-12-09 · Filed 2011-08-11 · Granted · Published 2011-11-23
The invention belongs to the **solar module** technical field, be specifically related to the transpa battery component of a kind of two-sided generating, be applicable to that large glass curtain parking lot, highway soundproof wall etc. are architectural, also be applicable to ...

Weatherproof building envelope

Tools


```
{
  "search_metadata": {
    "id": "695926f58c24bd247f1be7c9",
    "status": "Success",
    "json_endpoint": "https://serpapi.com/searches/acdcb
    "created_at": "2026-01-03 14:25:57 UTC",
    "processed_at": "2026-01-03 14:25:57 UTC",
    "google_patents_url": "https://patents.google.com/xh
    "raw_html_file": "https://serpapi.com/searches/acdcb
    "prettify_html_file": "https://serpapi.com/searches/
    "total_time_taken": 0.92
  },
  "search_parameters": {
    "engine": "google_patents",
    "q": "solar panel"
  },
  "search_information": {
    "total_results": 118864,
    "total_pages": 100,
    "page_number": 1
  },
  "organic_results": [
    {
      "position": 1,
      "rank": 0,
      "patent_id": "patent/EP2264377A2/en",
      "patent_link": "https://patents.google.com/patent/
      "serpapi_link": "https://serpapi.com/search.json?e
      "title": "Apparatus for two-way tracing and conc
      "snippet": "a plurality of second open push rods
```

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Flat roof photovoltaic foundation support system

CN • CN212752199U • 杨建 • 湖南馨雅林工程技术有限公司

Priority 2020-07-17 • Filed 2020-07-17 • Granted 2021-03-19 • Published 2021-03-19

The utility model belongs to the photovoltaic basic support field, in particular to a flat roof pho support system, aiming at the problems that the angle can not be adjusted to adapt to differer conditions generally in the operation process of the prior **solar panel**, the ...

Foldable photovoltaic module

CN • CN218217249U • 宋睿 • 夏能科技(北京)有限公司

Priority 2022-06-28 • Filed 2022-06-28 • Granted 2023-01-03 • Published 2023-01-03

The utility model discloses a collapsible photovoltaic module, the on-line screen storage devic base, base lower extreme fixed mounting has three pulley, base upper end fixed mounting has mechanism, rotary mechanism upper end fixed mounting has the connection pad, the equal ...

Next About 118864 results


```
"inventor": "Young Hwan Park",
"assignee": "Green Plus Co. Ltd.",
"publication_number": "EP2264377A2",
"language": "en",
"thumbnail": "https://patentimages.storage.googlea
"pdf": "https://patentimages.storage.googleapis.co
"figures": [
  {
    "thumbnail": "https://patentimages.storage.goo
    "full": "https://patentimages.storage.googlea
  },
  {
    "thumbnail": "https://patentimages.storage.goo
    "full": "https://patentimages.storage.googlea
```


GET

```
https://serpapi.com/search.json?engine=google_patents&q=solar+panel
```

Code to integrate Script Go PHP Java Rust .NET Google Sheets

```
require "serpapi"

client = SerpApi::Client.new(
  engine: "google_patents",
  q: "solar panel",
  api_key: "secret_api_key"
)

results = client.search
organic_results = results[:organic_results]
```

Free Plan · 250 searches / month

Get started

API Parameters

Search Query

q Optional

Parameter defines the query you want to search. You can split multiple search terms with semicolon ; . For advanced search syntax, please refer to About Google Patents.

Example for multiple search terms (separated by semicolon `;`):
`(Coffee) OR (Tea);(A47J)`

Pagination

page — Optional

Parameter defines the page number. It's used for pagination. (e.g., `1` (default) is the first page of results, `2` is the 2nd page of results, etc.).

num — Optional

Parameter controls the number of results per page. Minimum: `10` , Maximum: `100` .

Advanced Google Patents Parameters

sort — Optional

Parameter defines the sorting method. By default, the results are sorted by Relevance.
List of supported values are:
`new` - Newest
`old` - Oldest
Patent results are sorted by `filing_date` while scholar results are sorted by `publication_date` for `new` and `old` values.

clustered — Optional

Parameter defines how the results should be grouped.
List of supported values are:
`true` - Classification

dups — Optional

Parameter defines the method of deduplication. Either Family (default) or Publication.
List of supported values are:
`language` - Publication

patents — Optional

Parameter controls whether or not to include Google Patents results. (Defaults to

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scholar Optional

Parameter controls whether or not to include Google Scholar results. (Defaults to false)

Date Range

before Optional

Parameter defines the maximum date of the results. The format of this field is `type:YYYYMMDD` . `type` can be one of `priority` , `filing` , and `publication` .
Example:
- `priority:20221231`
- `publication:20230101`

after Optional

Parameter defines the minimum date of the results. The format of this field is `type:YYYYMMDD` . `type` can be one of `priority` , `filing` , and `publication` .
Example:
- `priority:20221231`
- `publication:20230101`

Participants

inventor Optional

Parameter defines the inventors of the patents. Split multiple inventors with `,` (comma). Wrap names containing commas in parentheses, e.g. `(Doe, John)` .

assignee Optional

Parameter defines the assignees of the patents. Split multiple assignees with `,` (comma). Wrap names containing commas in parentheses, e.g. `(Tesla, Inc)` .

Advanced Filters

country Optional

Parameter filters patent results by countries. Split multiple country codes with `,` (comma)

language Optional

Parameter filters patent results by languages. Split multiple languages with `,` (comma).
List of supported values are:
`ENGLISH`, `GERMAN`, `CHINESE`, `FRENCH`, `SPANISH`, `ARABIC`, `JAPANESE`, `KOREAN`, `PORTUGUESE`, `RUSSIAN`, `ITALIAN`, `DUTCH`, `SWEDISH`, `FINNISH`, `NORWEGIAN`, `DANISH`.
Example: `ENGLISH,GERMAN`.

status Optional

Parameter filters patent results by status.
List of supported values are:
`GRANT` - Grant
`APPLICATION` - Application

type Optional

Parameter filters patent results by type.
List of supported values are:
`PATENT` - Patent
`DESIGN` - Design

litigation Optional

Parameter filters patent results by litigation status.
List of supported values are:
`YES` - Has Related Litigation
`NO` - No Known Litigation

Serpapi Parameters

engine Required

Set parameter to `google_patents` to use the Google Patents API engine.

no_cache Optional

Parameter will force SerpApi to fetch the Google Patents results even if a cached version is already present. A cache is served only if the query and all parameters are exactly the same. Cache expires after 1h. Cached searches are free, and are not counted towards your searches per month. It can be set to `false` (default) to allow results from the cache, or `true` to disallow results from the cache. no_cache and

async Optional

Parameter defines the way you want to submit your search to SerpApi. It can be set to `false` (default) to open an HTTP connection and keep it open until you got your search results, or `true` to just submit your search to SerpApi and retrieve them later. In this case, you'll need to use our Searches Archive API to retrieve your results. `async` and `no_cache` parameters should not be used together. `async` should not be used on accounts with Ludicrous Speed enabled.

zero_trace Optional

Enterprise only. Parameter enables ZeroTrace mode. It can be set to `false` (default) or `true`. Enable this mode to skip storing search parameters, search files, and search metadata on our servers. This may make debugging more difficult.

api_key Required

Parameter defines the SerpApi private key to use.

output Optional

Parameter defines the final output you want. It can be set to json (default) to get a structured `JSON` of the results, or `html` to get the raw html retrieved.

json_restrictor Optional

Parameter defines the fields you want to restrict in the outputs for smaller, faster responses. See JSON Restrictor for more details.

API Results

JSON Results

JSON output includes structured data for Organic Results and Summary.

A search status is accessible through `search_metadata.status`. It flows this way: `Processing` -> `Success` || `Error`. If a search has failed, `error` will contain an error message. `search_metadata.id` is the search ID inside SerpApi.

HTML Results

This API does not have html response, just a text.

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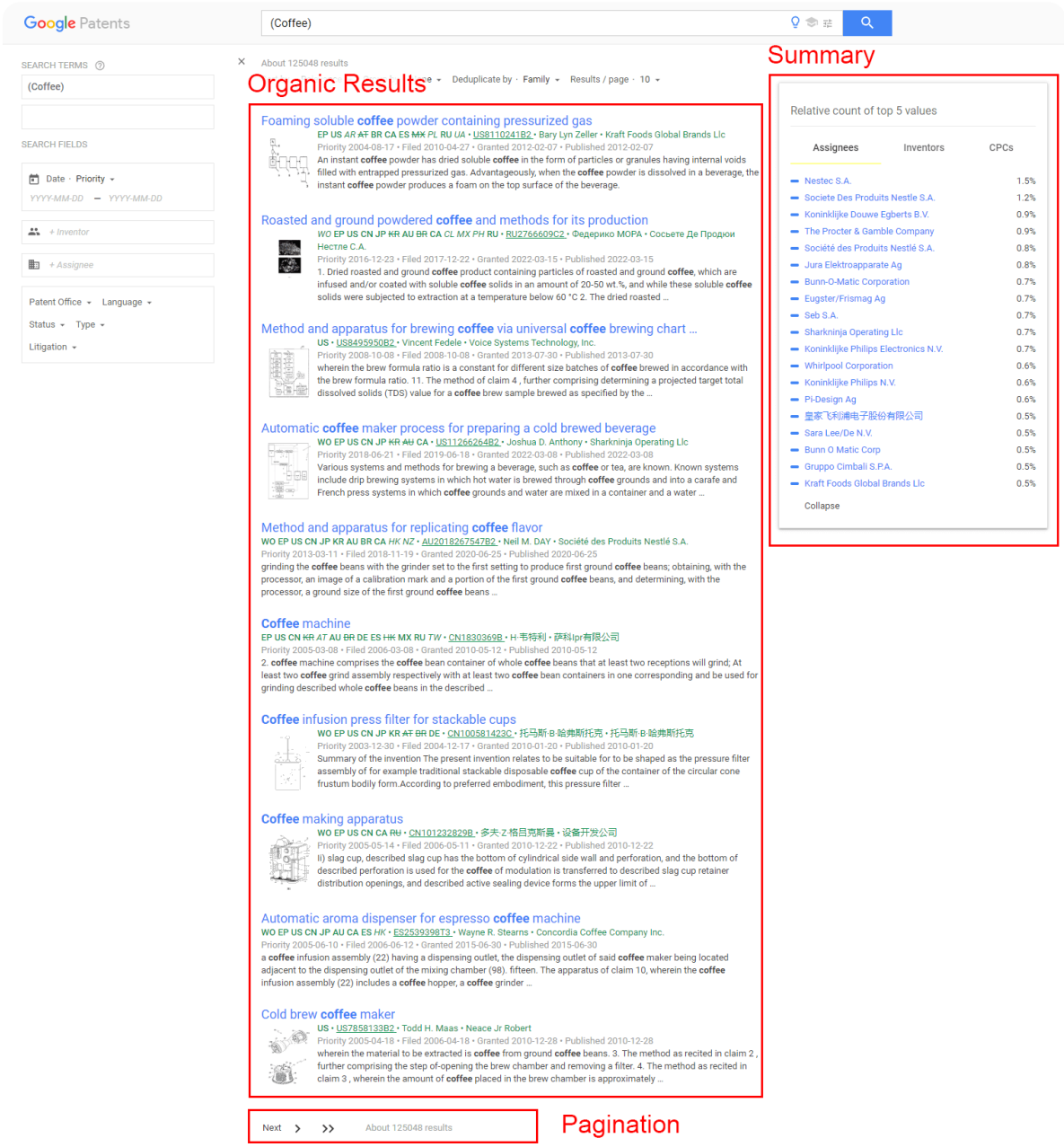
API Examples

Example with `q`: `(Coffee)` parameter



GET

```
https://serpapi.com/search.json?engine=google_patents&q=(Coffee)
```

Code to integrate Ruby

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```
  engine: "google_patents",
  q: "(Coffee)",
  api_key: "secret_api_key"
)

results = client.search
organic_results = results[:organic_results]
```

JSON Example


```
{
  "search_metadata": {
    "id": "64afe2d915afff17e18d922f",
    "status": "Success",
    "json_endpoint": "https://serpapi.com/searches/dadfa7408bfa2d66/64afe2d915afff17e18d9
    "created_at": "2023-07-13 11:41:13 UTC",
    "processed_at": "2023-07-13 11:41:13 UTC",
    "google_patents_url": "https://patents.google.com/xhr/query?url=q%3D%2528Coffee%2529"
    "raw_html_file": "https://serpapi.com/searches/dadfa7408bfa2d66/64afe2d915afff17e18d9
    "prettify_html_file": "https://serpapi.com/searches/dadfa7408bfa2d66/64afe2d915afff17
    "total_time_taken": 1.09
  },
  "search_parameters": {
    "engine": "google_patents",
    "q": "(Coffee)"
  },
  "search_information": {
    "total_results": 125048,
    "total_pages": 100,
    "page_number": 0
  },
  "organic_results": [
    {
      "position": 1,
      "rank": 0,
      "patent_id": "patent/US8110241B2/en",
      "patent_link": "https://patents.google.com/patent/US8110241B2/en",
      "serpapi_link": "https://serpapi.com/search.json?engine=google_patents_details&pate
      "title": "Foaming soluble coffee powder containing pressurized gas",
      "snippet": "An instant coffee powder has dried soluble coffee in the form of partic
      "priority_date": "2004-08-17",
      "filing_date": "2010-04-27",
      "grant_date": "2012-02-07",
      "publication_date": "2012-02-07",
      "inventor": "Bary Lyn Zeller",
      "assignee": "Kraft Foods Global Brands Llc",
      "publication_number": "US8110241B2",
      "language": "en",
      "thumbnail": "https://patentimages.storage.googleapis.com/25/a4/3c/08a69758ca974c/U
      "pdf": "https://patentimages.storage.googleapis.com/86/1c/bd/89a41d9faca215/US81102
      "figures": [
        {
```

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```
      {
        "thumbnail": "https://patentimages.storage.googleapis.com/c1/91/68/a0fc2ab8594€
        "full": "https://patentimages.storage.googleapis.com/5d/b3/37/ccdcbe58cbac2e/US
      }
    ],
    "country_status": {
      "EP": "ACTIVE",
      "US": "ACTIVE",
      "AR": "UNKNOWN",
      "AT": "NOT_ACTIVE",
      ...
    }
  },
  {
    "position": 2,
    "rank": 1,
    "patent_id": "patent/RU2766609C2/en",
    "patent_link": "https://patents.google.com/patent/RU2766609C2/en",
    "serpapi_link": "https://serpapi.com/search.json?engine=google_patents_details&pate
    "title": "Roasted and ground powdered coffee and methods for its production",
    "snippet": "1. Dried roasted and ground coffee product containing particles of roas
    "priority_date": "2016-12-23",
    "filing_date": "2017-12-22",
    "grant_date": "2022-03-15",
    "publication_date": "2022-03-15",
    "inventor": "Федерико МОРА",
    "assignee": "Сосьете Де Продюи Нестле С.А.",
    "publication_number": "RU2766609C2",
    "language": "en",
    "thumbnail": "https://patentimages.storage.googleapis.com/23/fc/61/43bb57210f8b7f/0
    "pdf": "https://patentimages.storage.googleapis.com/ab/3c/bb/d5d4b21616f08c/RU2766€
    "figures": [
      {
        "thumbnail": "https://patentimages.storage.googleapis.com/e1/a2/76/38d9b190f71e
        "full": "https://patentimages.storage.googleapis.com/d1/86/b8/556e6f1e4d609b/00
      }
    ],
    "country_status": {
      "WO": "UNKNOWN",
      "EP": "ACTIVE",
      "US": "ACTIVE",
      ...
    }
  },
  {
    "position": 3,
    "rank": 2,
    "patent_id": "patent/US8495950B2/en",
    "patent_link": "https://patents.google.com/patent/US8495950B2/en",
    "serpapi_link": "https://serpapi.com/search.json?engine=google_patents_details&pate
    "title": "Method and apparatus for brewing coffee via universal coffee brewing char
    "snippet": "wherein the brew formula ratio is a constant for different size batches
    "priority_date": "2008-10-08",
    "filing_date": "2008-10-08",
    "grant_date": "2013-07-30",
    "publication_date": "2013-07-30"
```

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```
      "language": "en",
      "thumbnail": "https://patentimages.storage.googleapis.com/f9/c6/73/311cd476a738f9/U
      "pdf": "https://patentimages.storage.googleapis.com/9b/13/3b/04ede26a15302b/US8495S
      "figures": [
        {
          "thumbnail": "https://patentimages.storage.googleapis.com/48/2a/2c/66f95a4e799:
          "full": "https://patentimages.storage.googleapis.com/68/0e/7b/399c0396b895eb/US
        },
        {
          "thumbnail": "https://patentimages.storage.googleapis.com/f8/75/c2/66c2c1a7fce8
          "full": "https://patentimages.storage.googleapis.com/97/31/80/9115dc57135085/US
        },
        {
          "thumbnail": "https://patentimages.storage.googleapis.com/38/05/98/b4b5eb0dfbee
          "full": "https://patentimages.storage.googleapis.com/69/f8/87/e7c19a808be041/US
        },
        ...
      ],
      "country_status": {
        "US": "ACTIVE"
      }
    },
    ...
  ],
  "summary": {
    "assignee": [
      {
        "key": "Total",
        "percentage": 100.0,
        "frequency": [
          {
            "year_range": "2022-2025",
            "percentage": 0.1
          },
          {
            "year_range": "2019-2022",
            "percentage": 2.7
          },
          ...
        ]
      },
       {
        "key": "Nestec S.A.",
        "percentage": 1.5,
        "frequency": [
          {
            "year_range": "2013-2016",
            "percentage": 0.1
          },
          {
            "year_range": "2010-2013",
            "percentage": 0.4
          },
          ...
        ]
```

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```
      "percentage": 1.2,
      "frequency": [
        {
          "year_range": "2019-2022",
          "percentage": 0.1
        },
        {
          "year_range": "2013-2016",
          "percentage": 0.2
        },
        ...
      ]
    },
    ...
  ],
  "inventor": [
    {
      "key": "Total",
      "percentage": 100.0,
      "frequency": [
        {
          "year_range": "2022-2025",
          "percentage": 0.1
        },
        {
          "year_range": "2019-2022",
          "percentage": 2.7
        },
        ...
      ]
    },
    {
      "key": "Ernesto Illy",
      "percentage": 0.6,
      "frequency": [
        {
          "year_range": "1989-1992",
          "percentage": 0.1
        },
        {
          "year_range": "1983-1986",
          "percentage": 0.1
        },
        ...
      ]
    },
    {
      "key": "Daniel Fischer",
      "percentage": 0.6,
      "frequency": [
        {
          "year_range": "2007-2010",
          "percentage": 0.1
        },
        {
          "year_range": "2004-2007",
```

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```
            ]
          },
          ...
        ],
        "cpc": [
          {
            "key": "Total",
            "percentage": 100.0,
            "frequency": [
              {
                "year_range": "2022-2025",
                "percentage": 0.1
              },
              {
                "year_range": "2019-2022",
                "percentage": 2.7
              },
              ...
            ]
          },
          {
            "key": "A47J",
            "percentage": 66.7,
            "frequency": [
              {
                "year_range": "2022-2025",
                "percentage": 0.1
              },
              {
                "year_range": "2019-2022",
                "percentage": 1.7
              },
              ...
            ]
          },
          {
            "key": "A23F",
            "percentage": 19.5,
            "frequency": [
              {
                "year_range": "2022-2025",
                "percentage": 0.1
              },
              {
                "year_range": "2019-2022",
                "percentage": 0.8
              },
              ...
            ]
          },
          ...
        ]
      },
      "pagination": {
        "current": 0,
```

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```
    "current": 0,
    "next": "https://serpapi.com/search.json?engine=google_patents&page=1&q=%28Coffee%29"
  }
}
```


JSON structure overview


```
{
  ...
  "organic_results": [
    {
      "position": "Integer - Position of the organic result",
      "rank": "Integer - Rank of the organic result - It may be different from `position`
      "cpc": "String - Cooperative Patent Classification (CPC) to the patent", // only pre
      "cpc_description": "String - The description of the CPC", // only present when `clus
      "title": "String - The title of the patent / scholar",
      "snippet": "String - The snippet of the patent / scholar",
      "publication_date": "String - The publication date of the patent / scholar",
      "is_scholar": "Boolean - True if it's a scholar (Google Scholar) result",
      "patent_id": "String - ID of the patent",
      "scholar_id": "String - ID of the scholar",
      "patent_link": "String - URL to the patent page",
      "scholar_link": "String - URL to the scholar page",
      "serpapi_link": "String - URL to SerpApi Google Patents Details API",
      // following fields are patent results only
      "priority_date": "String - The priority date",
      "filing_date": "String - The filed date",
      "grant_date": "String - The granted date",
      "inventor": "String - Name of the inventor",
      "assignee": "String - Name of the assignee",
      "publication_number": "String - Publication number of the patent",
      "language": "String - Language of the patent",
      "thumbnail": "String - URL to the thumbnail of the patent",
      "pdf": "String - URL to the PDF document of the patent",
      "figures": "Array - URLs to more images of the patent, normally includes a `thumbnai
      "country_status": "Hash - Statuses of the patent in each country. The key is the cou
      // following fields are scholar results only
      "url_hostname": "String - Hostname of the scholar's URL",
      "author": "String - Name of the author",
      "author_etal": "Boolean - Whether this scholar has three or more authors",
      "publication_venue": "String - Venue of publication",
    }
  ],
  "summary": {
    "assignee": [
      "key": "String - Name of the assignee",
      "percentage": "String - Percentage of the assignee",
      "frequency": [
        {
          "year_range": "String - Year range of the item",
```

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```
      ],
      "inventor": "Array - Summary of the inventors, the structure is the same as `assignee`
      "cpc": "Array - Summary of the CPCs, the structure is the same as `assignee`",
    },
    ...
  }
```

Api Status · GitHub · Legal · Security · Libraries · Release Notes · Public Roadmap · Support

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